73

Some causes of disease

A healthy person is one whose mental, physical, and chemical processes are working efficiently and harmoniously. Many things can upset this harmony and cause 'dis-ease'.

Organisms which cause disease

Diseases are often caused by organisms which live as **parasites**. A parasite is an organism which obtains its food from the living body of another organism called the **host**; but not all parasites harm their host. Most humans, for example, have a parasitic bacterium, *Escherischia coli*, living harmlessly in their intestines. Parasites which harm their hosts are said to be **pathogenic**, or **pathogens**.

Pathogenic micro-organisms

The words **micro-organism** and **microbe** refer to microscopic creatures including viruses, bacteria, rickettsias, protozoa, and fungi such as yeasts and moulds. The majority of microbes are harmless, and some are very useful, such as the yeasts used in baking and wine-making and the bacteria and fungi used to make antibiotics.

Pathogenic microbes are commonly known as germs. Examples are: viruses which cause influenza; bacteria which cause food poisoning and cholera; protozoa which cause dysentery and malaria; and fungi which cause ringworm and athlete's foot. Some of these microbes are described in Unit 74.

Parasitic worms and insects

Roundworms (nematodes) and flatworms (platyhelminths) which live as parasites are described in Units 75 and 76. Insects such as fleas and lice (described in Unit 77) can live as parasites but although they cause discomfort they are far more important as the carriers, or **vectors**, of pathogenic microbes. Female anopheline mosquitoes, for instance, are vectors of the microbes which cause malaria.

How pathogens cause disease

Figure 1 describes some of the ways in which pathogenic organisms enter the body. Once inside, some pathogens release poisonous chemicals called **toxins**. Most toxins are proteins and are by-products of the parasites' metabolism. They produce disease symptoms in the host like high temperature, headache, and vomiting. Symptoms do not appear immediately a pathogen enters a host. There is an interval called the **incubation period** before symptoms appear, during which germs multiply rapidly or larger parasites develop to full size.

Some parasites, including certain nematode worms and insects, bore through host tissues causing wounds which may then become infected by bacteria and viruses. This is called **secondary infection**.

Epidemic and endemic diseases

When a disease is always present in an area, such as malaria in parts of Africa, it is said to be **endemic** to that region. A sudden outbreak of the disease which spreads through a population is called an **epidemic**, and if it spreads through several countries the disease has become **pandemic**.

Notifiable diseases

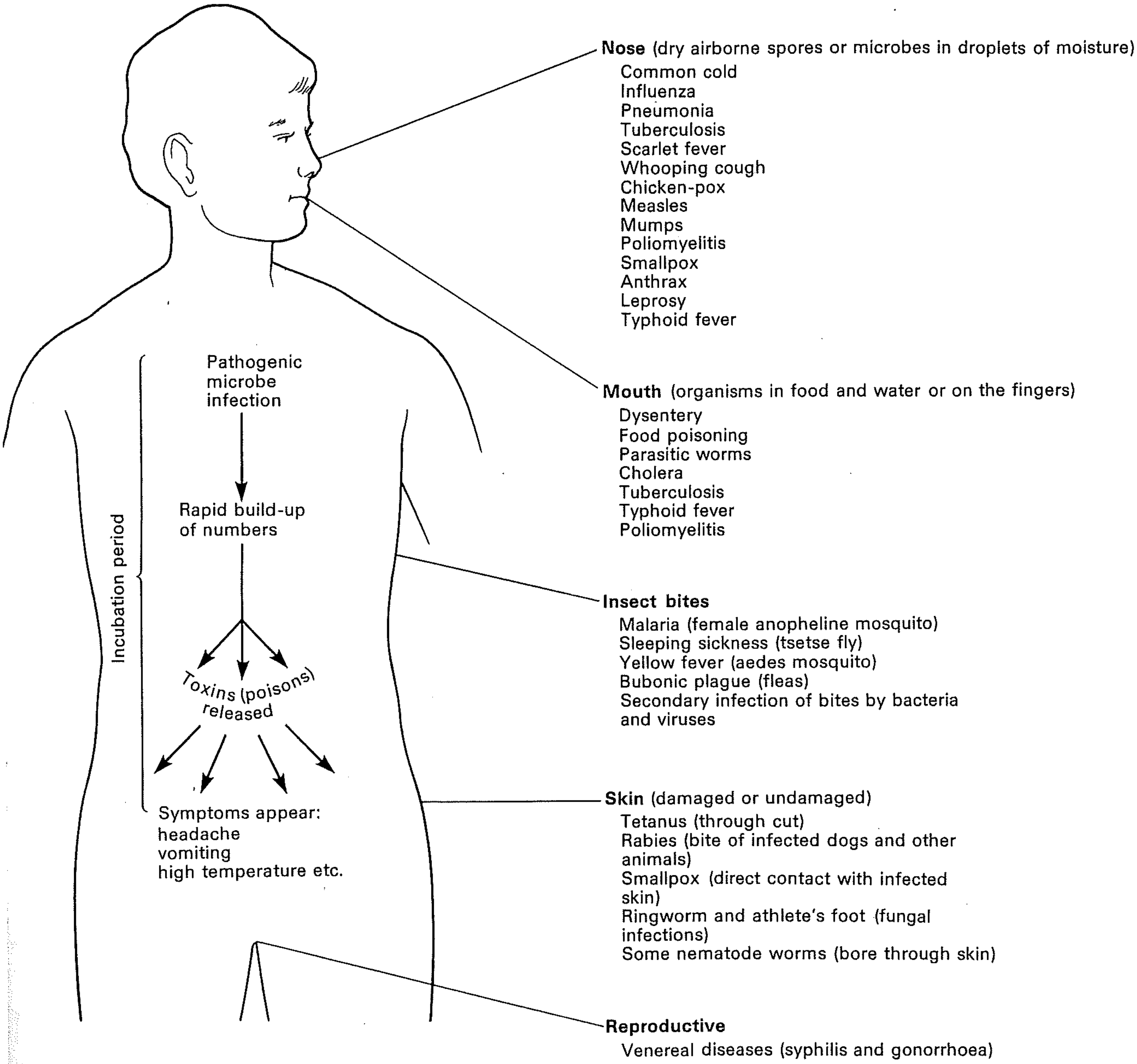
Doctors are compelled to report an outbreak of certain diseases to their Local Health Authority. These are called **notifiable diseases** and include: smallpox, scarlet fever, diphtheria, and typhoid fever. After notification, arrangements are made to prevent the spread of infection. Patients are isolated and their relatives, friends, and others who have been in contact with them are vaccinated.

People who have been in contact with an infectious disease are kept in **quarantine** for a time. This means they are isolated in a hospital for a few days longer than the incubation period of the disease to reduce risk of spreading the infection.

Other causes of disease

Parasites are not the only cause of disease. Many diseases such as haemophilia (described in Unit 36) are inherited. Diabetes results from defects in certain chemical processes in the body; and some mental illnesses, like anxiety states, can develop in people under great emotional stress.

Disease is also caused by a poor diet. Vitamin and mineral deficiency diseases are discussed in Unit 21. **Kwashiorkor** is a disease caused by lack of protein, which results in cracked skin and damage to the liver. Lack of fibre (bran) in the diet does not cause a deficiency disease but can contribute to the development of bowel cancer, piles, appendicitis, and constipation. There is also evidence that excess animal fats in the diet is one cause of **coronary heart disease**, a condition which can lead to heart attacks.



g. 1. How some disease-causing organisms enter the ody.

Some microbes which cause disease

Viruses and disease

Viruses are extremely small. A row of one million average-size viruses would be only 5 mm long. It is not clear whether viruses are alive or dead since they can be dried into a powder, stored for years in a bottle, and yet come to 'life' again when dissolved in water and placed on living tissue.

Viruses are not cells. They consist of DNA enclosed in an envelope of protein and fat. All viruses are parasites, only showing signs of life when inside their host cells. Here they reproduce as illustrated in Figure 1.

Diseases caused by viruses include chicken-pox, smallpox, common cold, influenza, measles, mumps, and rabies.

Bacteria and disease

Bacterial cells vary in size between 0.0005 mm and 0.005 mm in length. Like other cells they contain DNA but it is not in the form of chromosomes. The cytoplasm of a bacterium is contained within a tough cell wall and this is often covered with slime. Some bacteria can move about by means of whip-like flagella (Fig. 2).

In favourable conditions bacteria reproduce by dividing in two, a process called **fission** (Fig. 3). Under unfavourable conditions some bacteria can form thick-walled **spores** which can survive for years as dust.

Diseases caused by bacteria include boils, food poisoning, whooping cough, cholera, diphtheria, and venereal disease.

Protozoa and disease

Protozoa are complex single-celled organisms which vary enormously in size and shape. Very few cause disease in humans. *Entamoeba histolytica* is a protozoan which causes amoebic dysentery (Fig. 4A). This disease is rare in Britain but quite common in southern Europe. The amoebae can live harmlessly in the large intestine, feeding on bacteria. Under certain circumstances they become parasitic, destroying cells lining the large intestine, and forming large ulcers.

Fungi and disease

Very few fungi are parasites of humans. One of the commonest fungal diseases is called **ringworm**, because it can produce a circular swelling on the skin. Ringworm fungi also attack the scalp, and the soft skin of the groins. Another very similar fungus attacks the soft skin of the feet, especially between the toes, causing a disease called **athlete's foot**. Fungal diseases are spread by airborne spores (Fig. 5), by contact with infected people and, in the case of athlete's foot, by infected floors and mats on which people walk bare-foot.

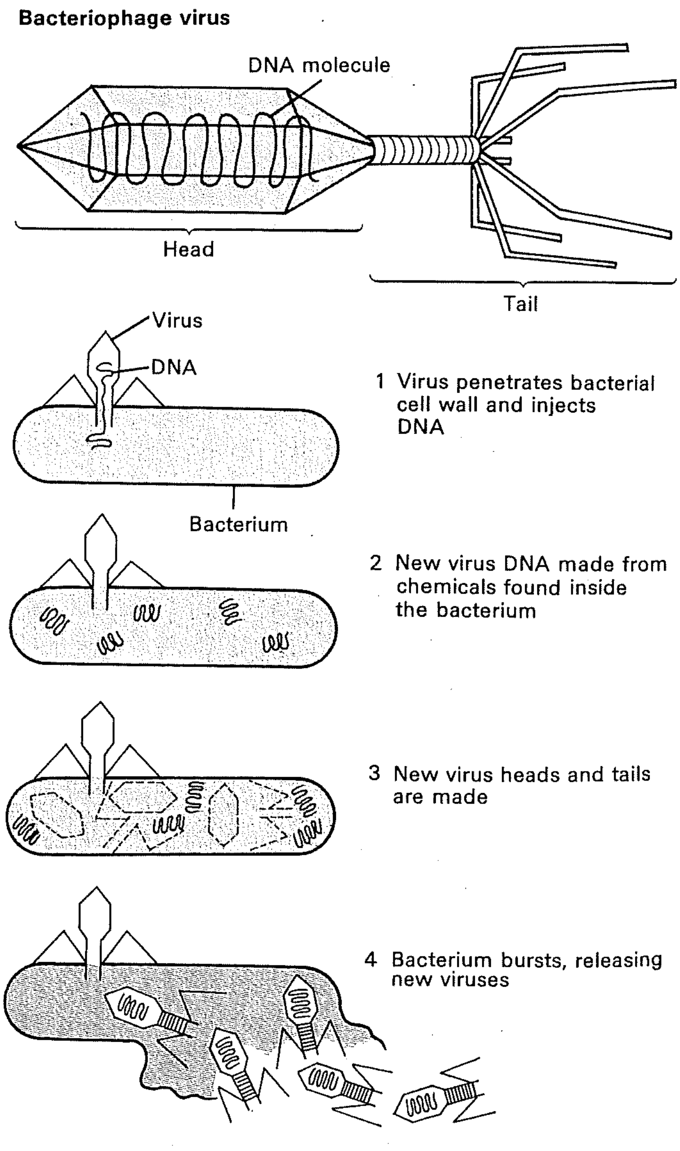
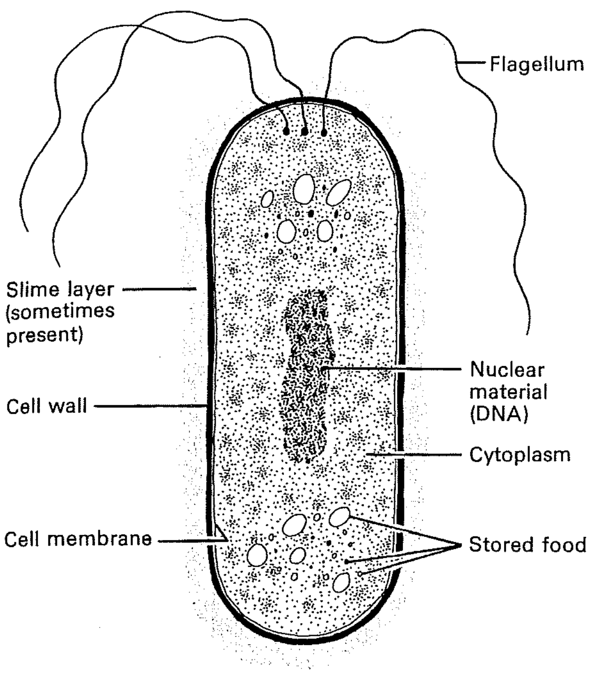


Fig. 1. How a virus reproduces. The virus illustrated here attacks bacteria and is called a bacteriophage.



ig. 2. **Diagram of a bacterial cell** showing most of the ructures found in bacteria.

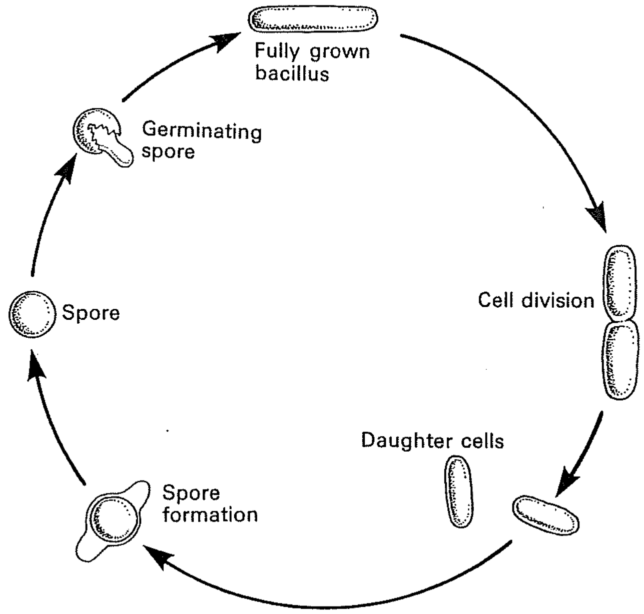


Fig. 3. Life-cycle of a spore-forming bacterium.

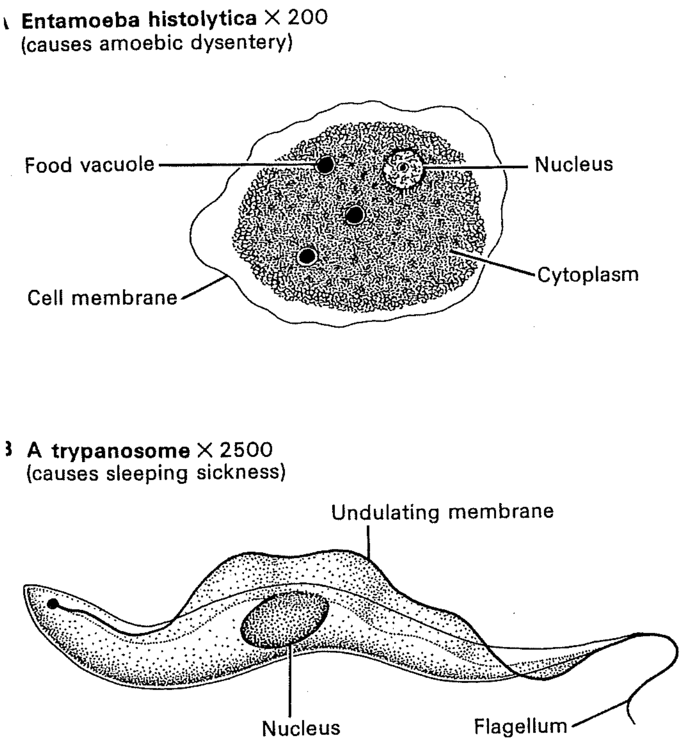


Fig. 4. **Some parasitic protozoa.**

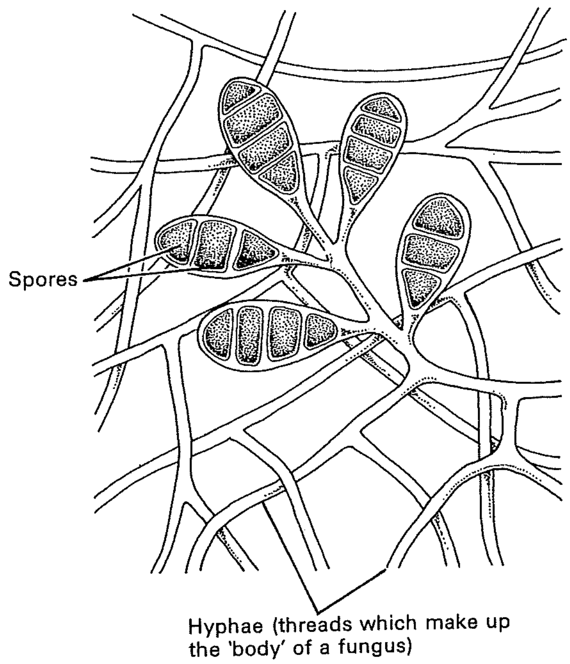


Fig. 5. **The spore-producing organs** of a microscopic fungus which causes ringworm.

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Parasitic roundworms (Nematodes)

Roundworms are round in cross-section and have long narrow cylindrical bodies, often pointed at both ends. They vary in size from microscopic to 30 cm long. Roundworms are covered with a smooth almost transparent cuticle. Their digestive system consists of a mouth, a muscular pharynx which sucks in food, and a fairly straight gut ending at an anus (Fig. 1). They have a simple nervous system, and muscles along the body which enable them to wriggle.

Roundworms live in soil, in fresh and salt water, and as parasites in almost every type of plant and animal. About 40 species are parasites of humans, one of the commonest being a roundworm of the genus *Ancylostoma*, commonly known as the hookworm.

Human hookworms

Hookworms cause much sickness in Italy and other parts of southern Europe, in the Middle and Far East, in North and South America and in South Australia. Hookworms, and all other roundworm parasites of humans, are very rare in Britain.

Adult hookworms are between 8 and 13 mm long. They live as parasites in the intestine where they suck blood by means of a bell-shaped mouth containing several teeth. The teeth are used to cut and tear the lining of the intestine until a blood vessel is opened. The wound is then injected with chemicals which prevent the blood clotting, so that the parasite receives a plentiful supply of food. Victims with a large number of hookworms can suffer severe anaemia.

Life history

Male and female worms mate, and soon afterwards the females release fertile eggs. These are passed out of the host in its faeces.

Further development depends upon the eggs coming to rest in warm damp soil. This can happen in regions with primitive sewage disposal arrangements, or where human excreta is used to fertilize crops. The eggs hatch releasing tiny larvae which feed on bacteria in the faeces and in surrounding soil. This **first larva** grows and moults its skin to become a **second larva**, which does the same to become a **third larva**. Only the third larva can infect a new host and so it is called the **infective stage** of the life-cycle. Third larvae can bore through thin skin on the feet, hands, and wrists of anyone who touches soil in which the larvae live. After penetrating the skin the larvae enter a blood vessel and are carried along by the bloodstream until they reach capillaries in the lungs. Here, they break out of the capillaries into the lungs and wriggle up the wind-pipe and down the gullet into the intestine. During this journey they become **fourth larvae**, which mature into adult males and females.

Control

Hookworm infection is best controlled by disposing of sewage in such a way that their eggs do not enter the soil.

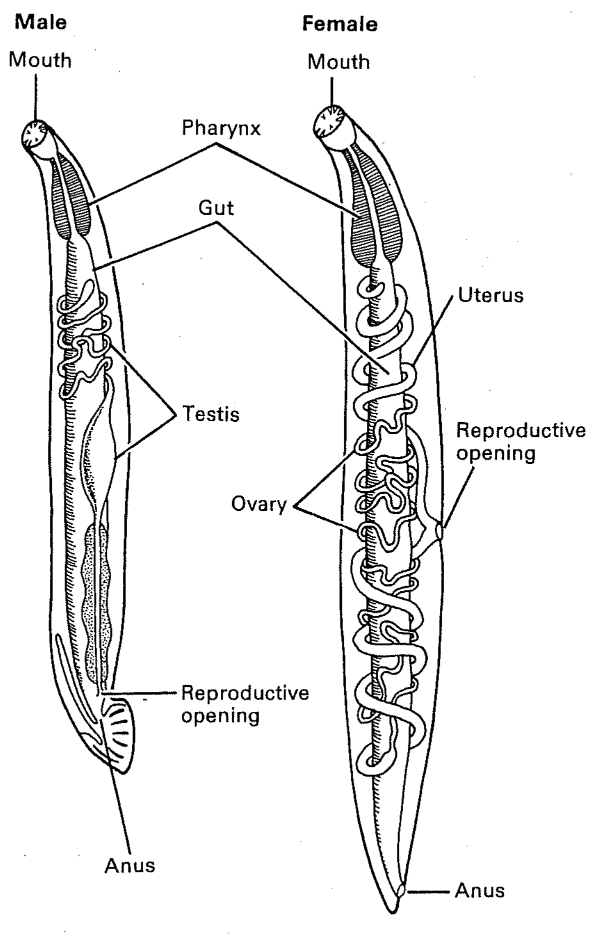

Fig. 1. Male and female hookworms. These are parasites which live in the human intestine where they suck blood.

5

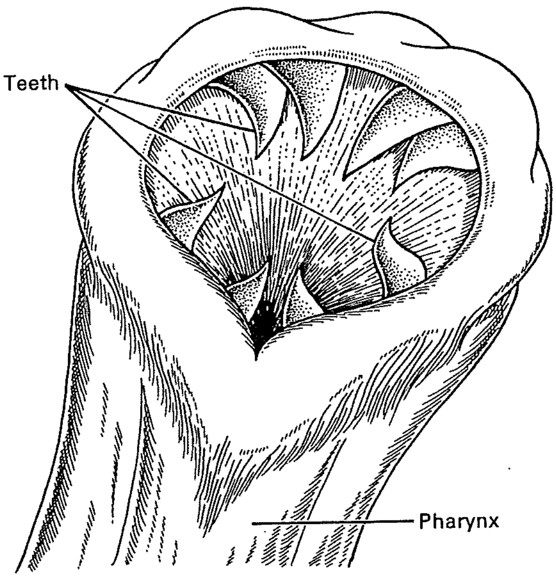

Fig. 2. Mouth of a hookworm.

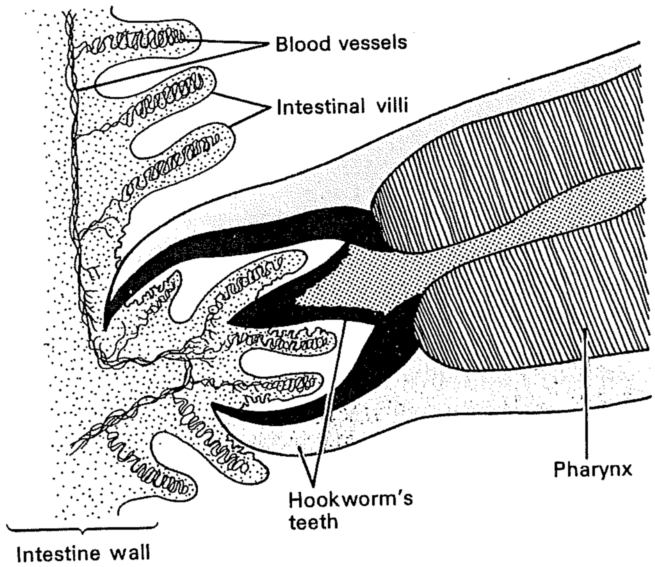

Fig. 3. A hookworm attached to gut wall with its teeth, sucking blood.

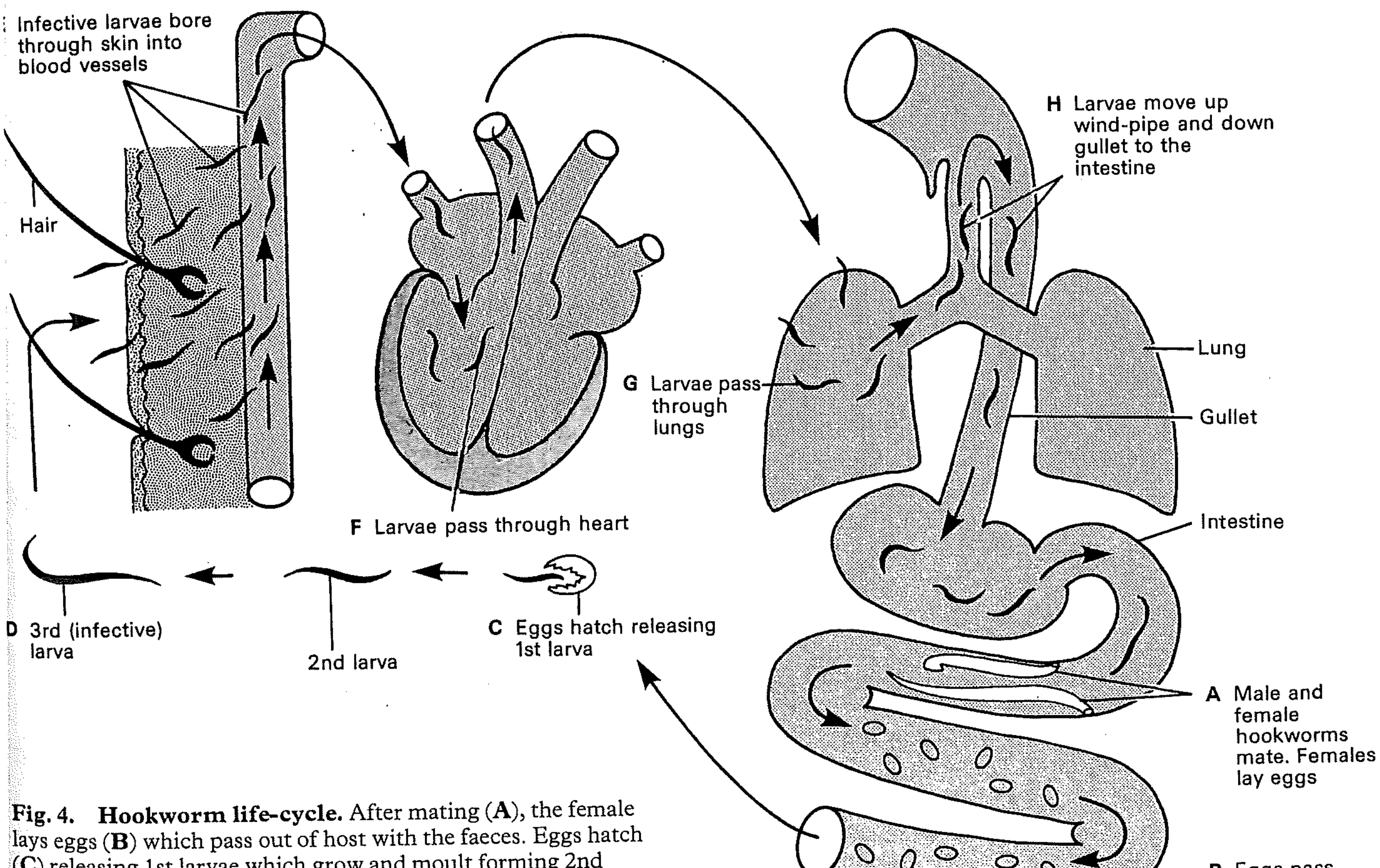

Fig. 4. Hookworm life-cycle. After mating (**A**), the female lays eggs (**B**) which pass out of host with the faeces. Eggs hatch (**C**) releasing 1st larvae which grow and moult forming 2nd larvae, which do the same forming 3rd larvae (**D**). These larvae infect new host by boring through the skin (**E**), and entering a blood vessel which transports them to the lungs (**F** and **G**). They climb up the wind-pipe and down the gullet (**H**) moulting on the way, and mature in intestine.

6

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The spread of infection

People can be infected with disease-causing organisms in many ways. The main sources of infection are: contact with infected people or with objects contaminated with germs; consumption of contaminated food and drink; bites from infected insects; and inhaling contaminated dust, or droplets of moisture breathed out by infected people.

Direct contact

Diseases spread by contact with an infected tissue of another person are said to be **contagious.** The venereal diseases gonorrhoea and syphilis can only be spread by direct physical contact because the bacteria which cause them die quickly outside the body.

Other diseases which can be spread by direct contact include fungal infections such as ringworm and athlete's foot; chicken-pox, smallpox, measles, leprosy; and boils and septic wounds.

A large number of germs can live for some time outside the body. If infected people leave such germs behind on objects which they touch, the objects then become sources of infection. This is known as spread of infection by indirect contact.

Indirect contact

The list of objects which can store germs is enormous. The most obvious are towels, coins, door handles, books, lavatory seats, a child's toys, and crockery, especially if cracked or chipped. Many of the diseases already mentioned are spread by indirect contact, especially fungal and other skin infections, and infections of the digestive and respiratory systems. It is dangerous to use the comb, towel, or bedding of an infected person. Further, since germs can be transferred from objects to the hands and then to the mouth, it is important to wash your hands before meals and after using the lavatory.

Contaminated food and drink

Food can be contaminated with disease-causing organisms by the unwashed hands of an infected person or by contact with unprotected septic wounds; by flies which have previously visited excreta and decomposing waste; by the excreta, saliva, hairs, etc. of wild rats and mice and various pets; and by contaminated water used to wash food containers or food itself. The germs of typhoid and paratyphoid fever, dysentery, and salmonella food poisoning are spread like this.

Unit 76 describes how tapeworm infections can result from eating undercooked meat containing bladderworms. Drinking water can be contaminated with faeces and urine if sewage disposal arrangements are primitive or inefficient. Cholera in particular is spread this way.

Milk and milk products such as cream can carry germs if careless dairy workers fail to wash their equipment thoroughly. Tuberculosis and brucellosis are spread to humans by the milk of infected cows.

Airborne infection

People can be infected with anthrax by breathing in dusty air containing dried spores of the bacteria which cause this disease. Airborne germs also come from the bedding, clothing, and handkerchiefs of infected people.

A sneeze can hurl 20 000 droplets of moisture containing viruses and bacteria a distance of more than four metres. Germ-laden droplets are also coughed and breathed out by infected people. Germs can float in the air for long periods and be inhaled by others. This is called **droplet infection** and is responsible for spreading colds, influenza, pneumonia, diphtheria, whooping cough, and tuberculosis.

Insects and other vectors

Houseflies gather germs on their feet, bodies, and in their digestive systems as explained in Unit 78. These germs are transferred to humans when the flies settle on food. Flies spread typhoid fever, typhus, cholera, dysentery, and tapeworm eggs.

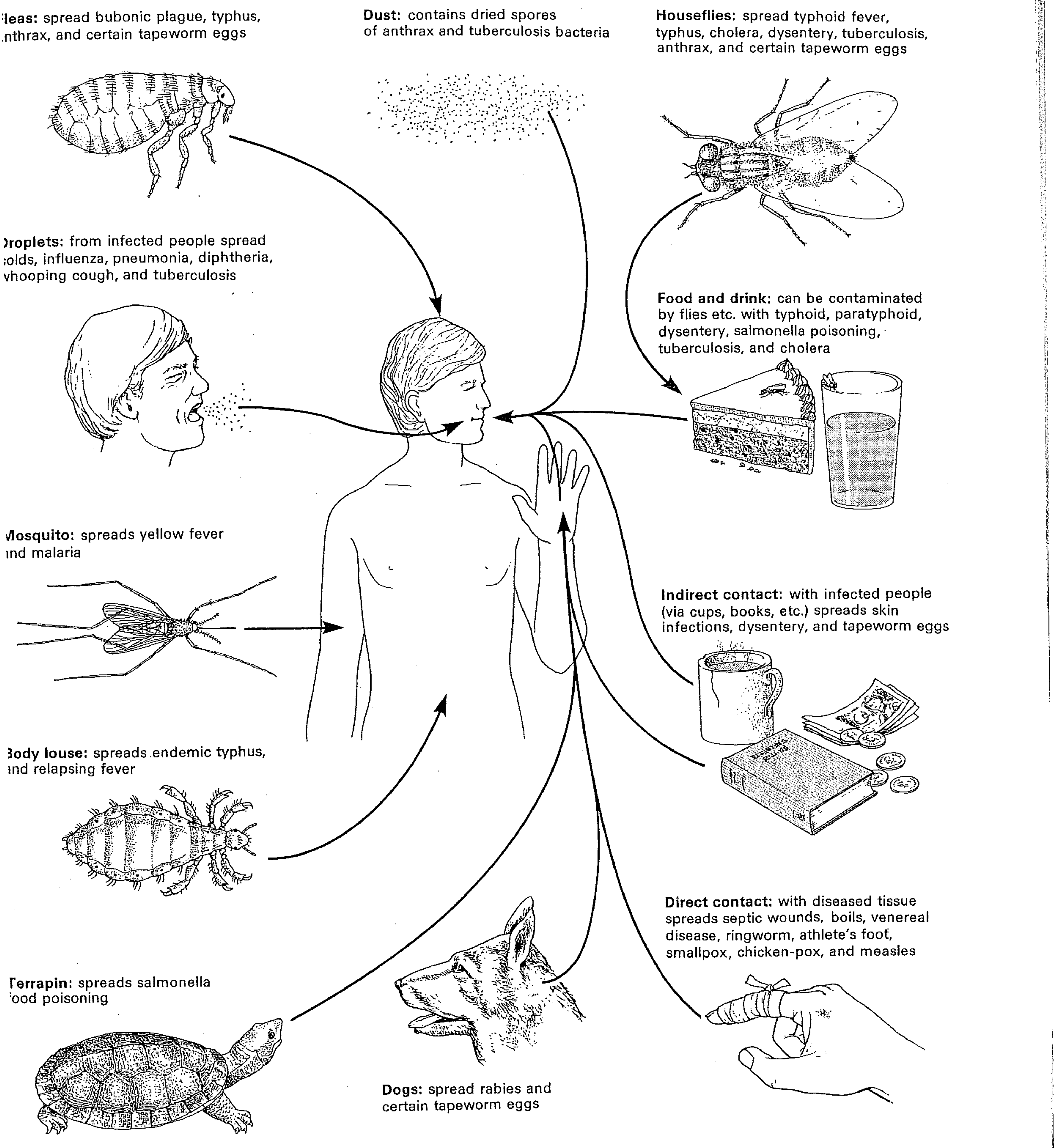
Blood-sucking insects transfer germs amongst humans and from animals to humans. Certain mosquitoes spread the protozoa which cause malaria and the yellow fever virus, and fleas transfer bubonic plague bacteria from rats to humans.

Rabies viruses are carried by dogs, foxes, and wolves. Infected animals become vicious and unpredictable and are likely to bite humans who come near, transferring the virus in their saliva.

Healthy carriers of disease

Healthy carriers are people who carry certain disease-causing organisms in their bodies without suffering any ill-effects. Consequently they are unaware of their condition and can spread germs to other people, causing mysterious outbreaks of disease.

7



ig. 1. **Some sources of infection.**

8

80

Prevention of infection

Good health can be maintained by keeping a good posture (Unit 17), regular exercise (Unit 18), eating a sensible diet (Unit 22), and by careful preservation of food (Unit 23).

Personal cleanliness is also very important. Care of the teeth is described in Unit 25, and Unit 44 describes care of the skin and hair. Figure 1 of this Unit summarizes how health can be maintained by personal cleanliness, and Figure 2 illustrates health hazards which can result from lack of hygiene in the home.

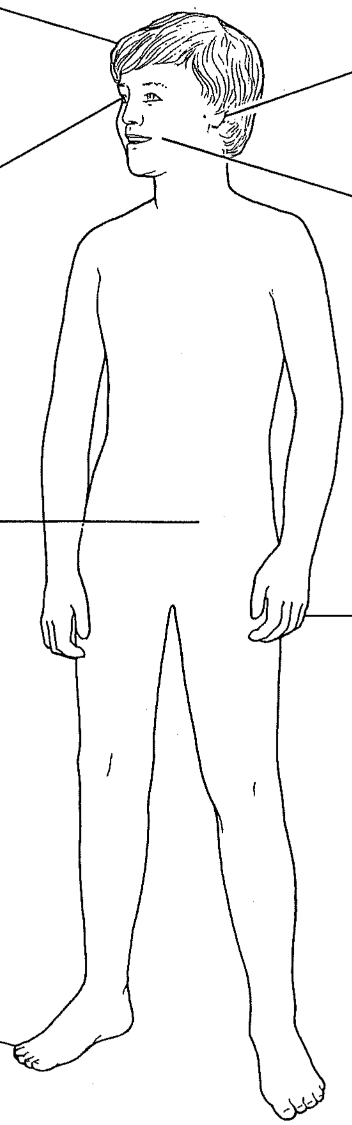


Fig. 1. Personal cleanliness is a very important part of the fight against infection.

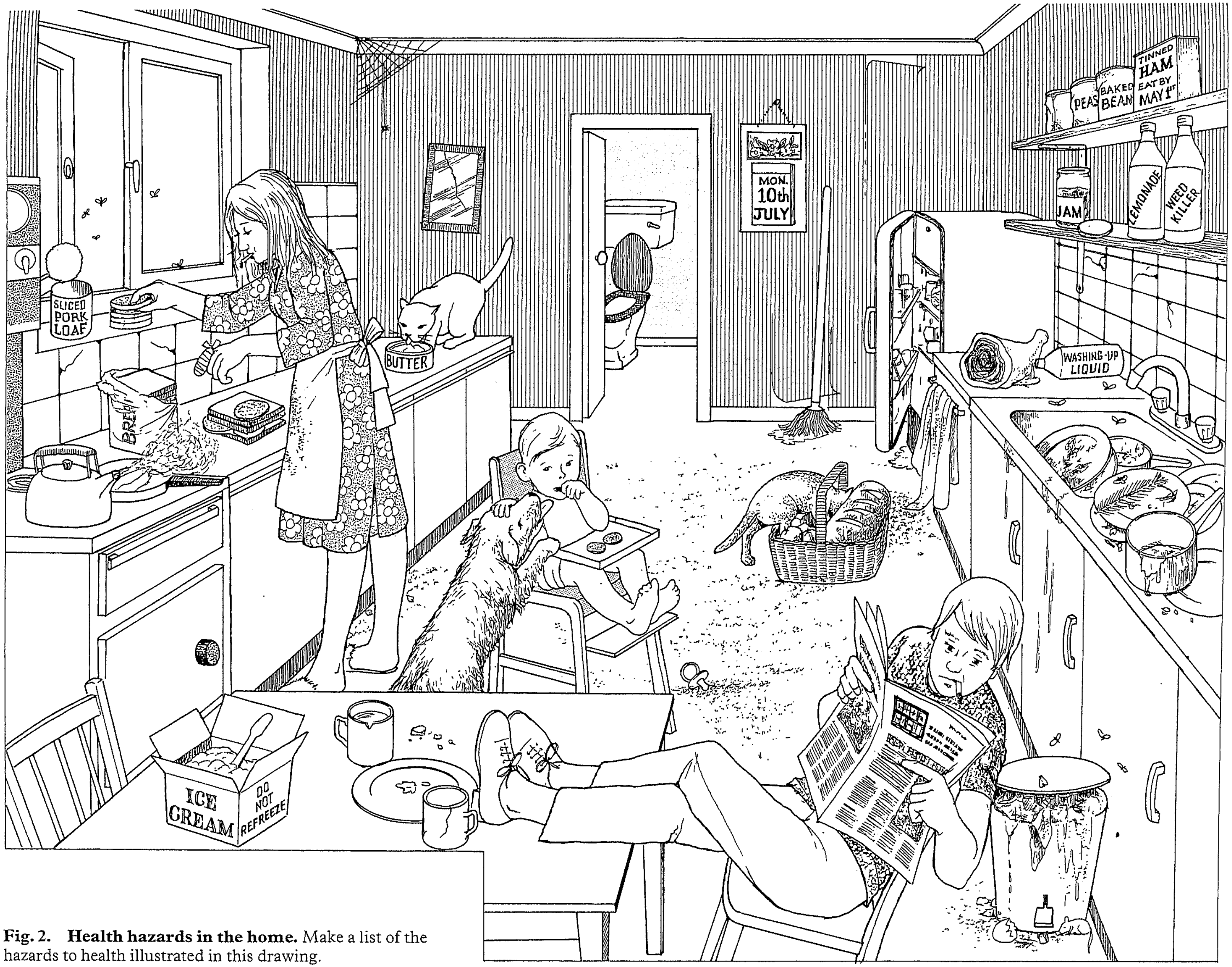


Fig. 2. Health hazards in the home. Make a list of the hazards to health illustrated in this drawing.

90

Natural immunity

The body possesses many natural defences against infection. These defences include the skin, the membranes covering the eyes and lining the respiratory system, secretions such as tears and digestive juices, and a type of white blood cell called a phagocyte. These are all examples of natural immunity.

The skin

The dead, flattened cells which form the outer layer of the skin form a physical barrier to the entry of germs into the body. As fast as these cells wear away they are replaced by more which grow from below, as described in Unit 43. The same cell growth repairs the skin when it is damaged.

This barrier of dead cells is kept supple and waterproof by an oily substance called **sebum** produced by the sebaceous glands (Fig. 1C). Sebum also contains an antiseptic chemical which kills some bacteria as they touch the skin.

The eyes

The eyes are protected from infection by a thin skin called the **conjunctiva**. This is continually bathed in an antiseptic liquid produced by the tear glands (Fig. 1A). Blinking spreads this liquid and washes away germs and dust from the eyes.

The respiratory system

The nasal passages, wind-pipe, and bronchial tubes of the lungs are lined with a carpet of microscopic hair-like structures called **cilia** (Fig. 1B). Cells between the cilia produce a sticky fluid called **mucus** which traps germs and dirt breathed in through the nose and mouth. Back-and-forth movements of cilia carries the mucus and trapped germs and dirt to the back of the throat where they are swallowed, rendered harmless by digestive juices, and passed out of the body in the faeces.

The digestive system

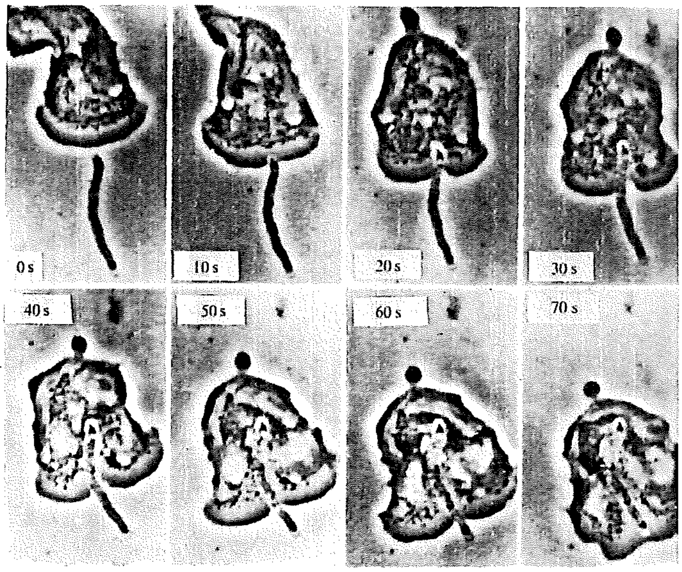
Many germs are unavoidably swallowed with food and drink. These germs are usually harmless but in any case the majority are killed by stomach acid (Fig. 1E).

Phagocytes

These are white blood cells which destroy bacteria that invade the body, by engulfing and digesting them. **Neurophils** and **monocytes** are examples of phagocytic cells (Fig. 1D).

Phagocytes are particularly active in wounds. Soon after the skin is damaged capillaries in surrounding tissues dilate, increasing the supply of blood to the area. Fluid pours out of these capillaries into the wound, together with thousands of phagocytes which destroy any bacteria present. Unit 36 describes how the wound is quickly sealed off by a blood clot.

Bacteria which penetrate deep into the body and enter the lymphatic system are killed by large phagocytes inside the lymph nodes, described in Unit 34. Bacteria inside the body are also killed by chemicals called antibodies, described in the next Unit.



White blood cells (phagocytes) destroying a bacterium.

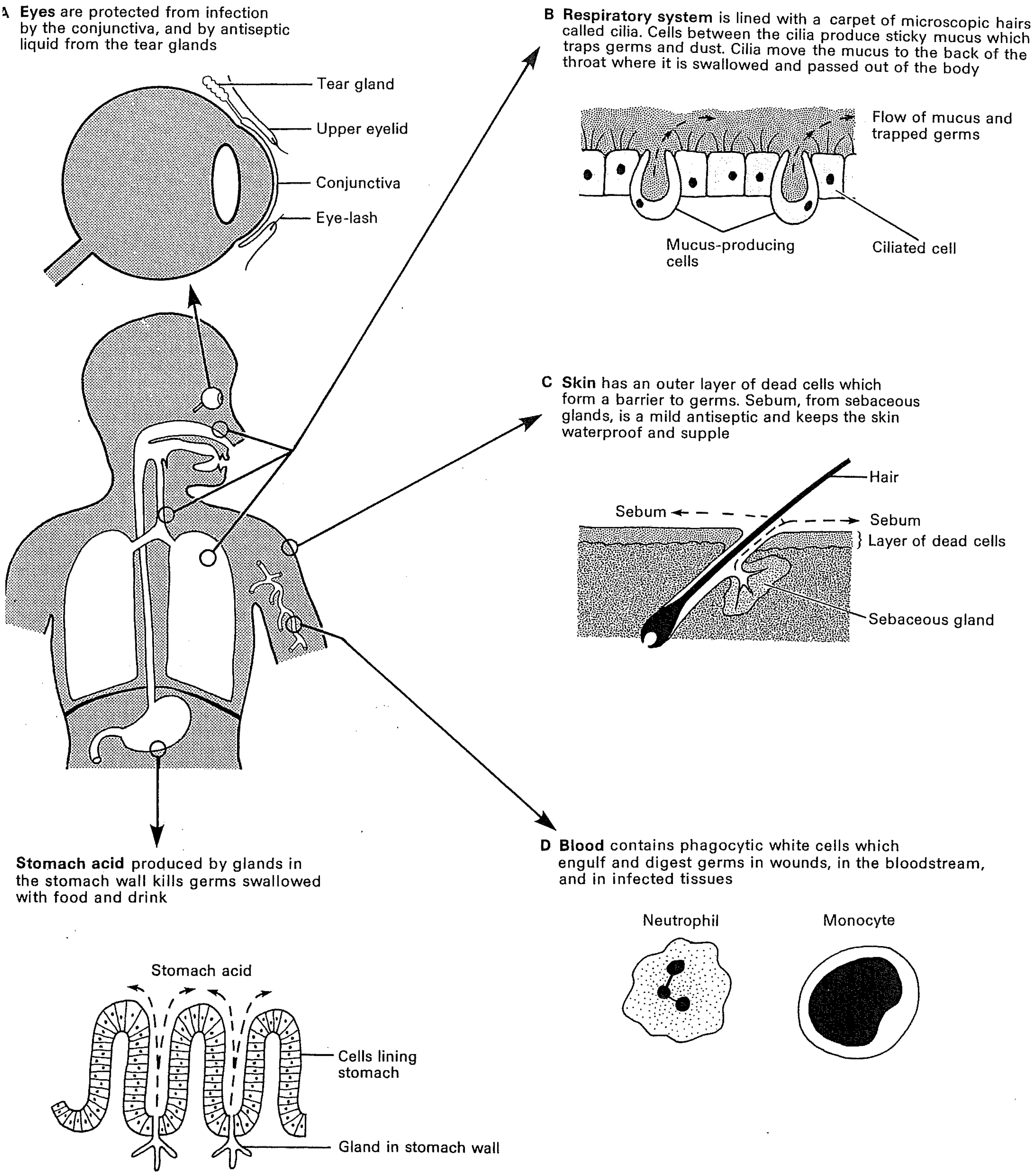


Fig. 1. The body's natural defences against infection.

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Acquired immunity and immunization

When germs penetrate the body's natural defences described in Unit 90, the body reacts by producing substances called **antibodies**. Antibodies circulate in the blood and tissue fluid killing germs or making them harmless. Antibodies also neutralize poisonous chemicals called **toxins** which germs produce.

Since antibodies appear in the body *after* an invasion by germs, their production is an example of **acquired immunity**. There are two types of acquired immunity: active and passive. Antibody formation is an example of **active immunity** because their formation is an active response to infection. Passive immunity is described later in this Unit.

Antibodies

Antibodies are proteins. They are made by white blood cells called lymphocytes, described in Units 29 and 34. Any substance which stimulates lymphocytes to make antibodies is called an **antigen**. Bacteria and viruses are covered with antigen molecules and toxins may also act as antigens.

When lymphocytes contact germs or toxins the antigen molecules are detected and antibodies are formed. Antibodies combine with the antigens producing a number of different effects.

1. **Opsonins** are antibodies which combine with antigen material on the outer surface of germs and appear to make the germs more likely to be destroyed by phagocytes (Fig. 1A). It is as if the antibody makes a germ more 'appetizing'.
2. **Lysins** are antibodies which kill germs by causing them to burst open into fragments which are engulfed by phagocytes (Fig. 1B).
3. **Agglutinins** are antibodies which cause germs to stick together in clumps (agglutinate). In this state the germs can neither penetrate cells nor reproduce properly (Fig. 1C).
4. **Anti-toxins** are antibodies which combine with toxins and render them harmless.

There are many different antigens and each requires a specific antibody to destroy it. The antibody which combines with measles virus antigen, for example, will destroy this virus and no other.

Antibodies are produced slowly at first but if a disease persists for a few days they are produced in much larger quantities. Moreover, the antibodies against certain germs remain in the body for many years after the infection has disappeared, thereby giving the body a built-in resistance to the germs if re-infection occurs. Should this happen, antibodies are produced much faster than during the first infection. A child who has recovered from chicken-pox, for instance, is unlikely to suffer from the disease again despite repeated exposure to infection.

Immunization

The body can be artificially stimulated into producing antibodies. This prepares it in advance to fight off infection. This is done by inoculating someone with **vaccine**. A vaccine is a liquid containing antigens powerful enough to stimulate antibody formation without causing harm. This is called **immunization** because it makes the body immune to germs with the same antigens as the vaccine.

One of the first people to use a vaccine with success was Edward Jenner (1749–1823). Jenner was investigating a theory that people who recovered from a mild disease called cowpox would, thereafter, be immune to smallpox, which is usually fatal. He scratched the skin of a healthy boy and rubbed pus from the hand of a girl with cowpox into the wound. The boy caught cowpox and when he had recovered Jenner inoculated him with pus from a smallpox victim. The boy did not catch smallpox. The modern explanation for this result is that cowpox and smallpox viruses have the same antigens. Thus if someone catches cowpox first they form antibodies which make them immune to smallpox. Jenner's method of immunization soon acquired the name **vaccination** after *vaccinia*, the Latin word for cowpox.

Modern vaccines contain either killed germs, germs made harmless to humans by growing them in non-human hosts, or germ-toxins made harmless in various ways. Vaccines are now available to combat many diseases including typhoid fever, poliomyelitis, cholera, and bubonic plague.

Passive immunity

It is now possible to inject people with ready-made antibodies in order to help them fight infection before their own antibody production has reached full speed. This treatment is called passive immunity: the body is protected without doing any work.

Tetanus antibodies, for example, are obtained by injecting horses with tetanus toxin. The horse makes antibodies to combat the toxin. Some blood is then removed from the horse, and serum, complete with tetanus antibodies, is extracted from it.

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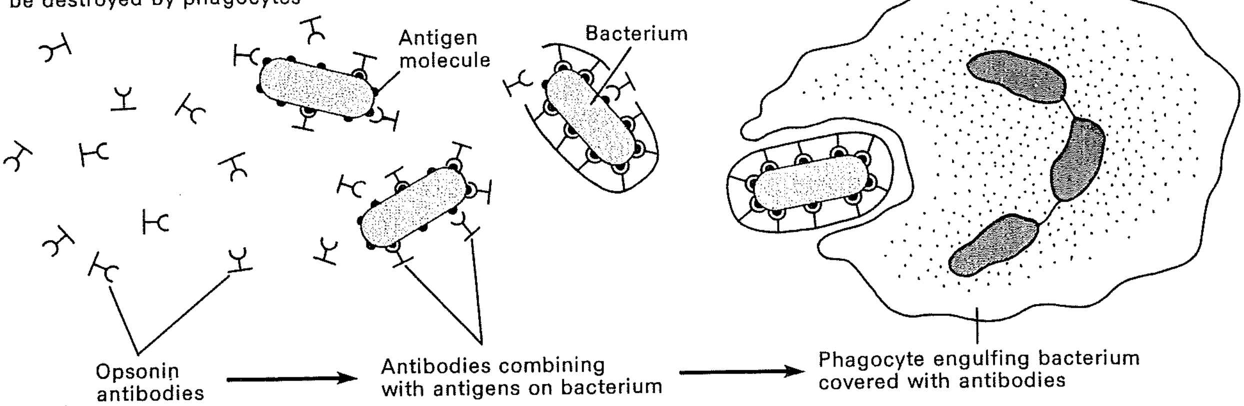
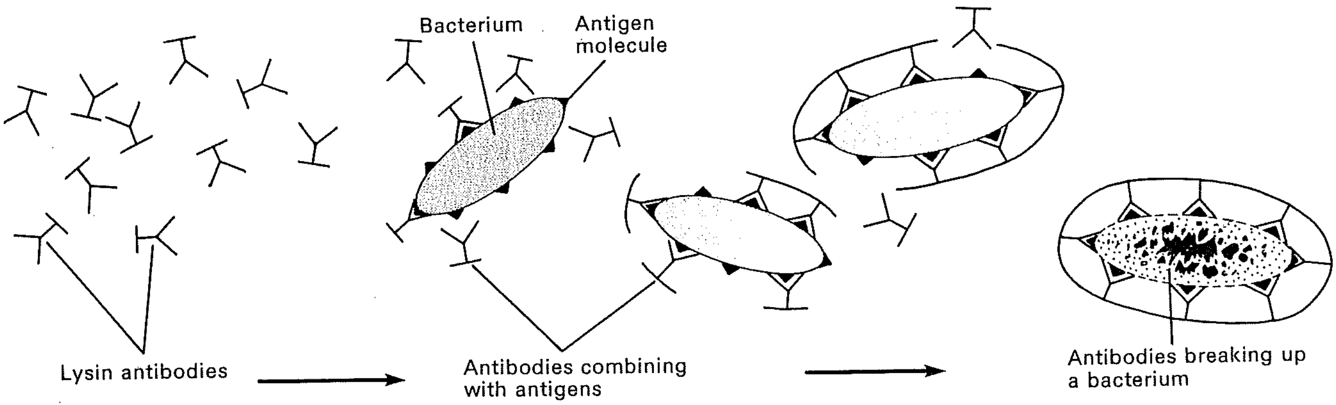
Opsonins are antibodies which combine with antigens on germs, making the germs more likely to be destroyed by phagocytes



Lysins are antibodies which break up germs into tiny fragments



C **Agglutinins** stick germs together so they cannot enter cells or reproduce

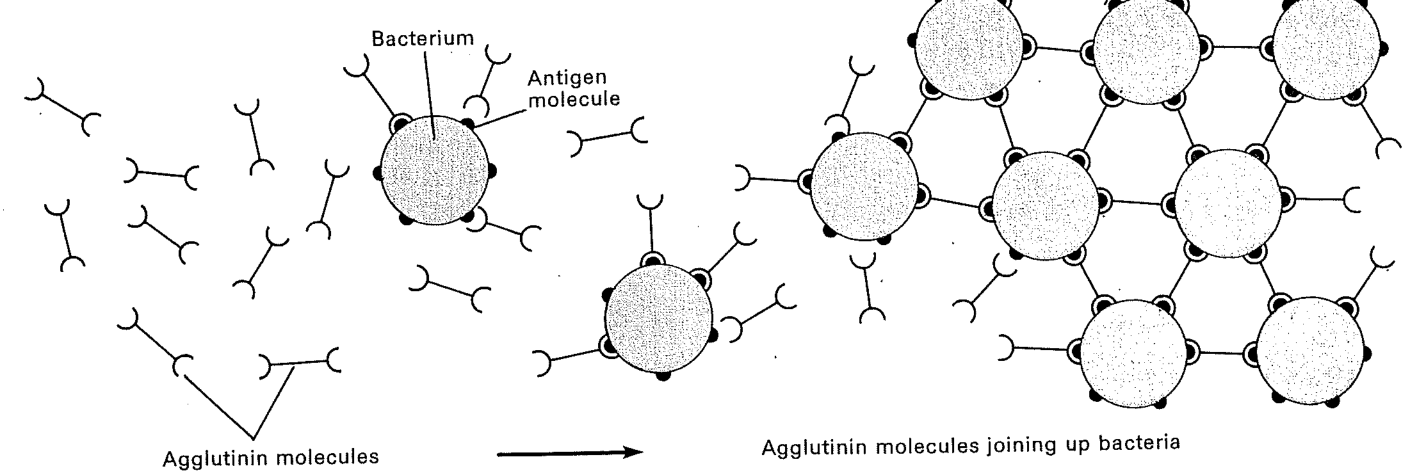


Fig. 1. Diagram of how antibodies destroy germs. Antibodies are made by white blood cells called lymphocytes to protect the body against germs. Each antibody combines with a specific antigen molecule on the germ surface. (Antibody molecules do not actually look like this. These shapes are used merely to indicate antibody–antigen reactions. Moreover, antibody molecules have been drawn much larger than life. In fact, on this scale they would be invisible.

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CHAPTER 8

Disease

8.1 Knowledge preview

Science understanding

FOUNDATION	STANDARD	ADVANCED

1 Diseases are conditions that can make you sick and stop your body from functioning normally. Diseases can be classified as infectious or non-infectious. Complete the following sentences by either writing 'infectious' or 'non-infectious'.

(a) Diseases that cannot be passed from one person to another are called ________

(b) Diseases that can be passed from one person to another are called ________

2 Classify the diseases in the box below as infectious or non-infectious by writing them in the correct column in the table.

AIDS	Alzheimer's disease	appendicitis	arthritis	athlete's foot (tinea)
breast cancer	chickenpox	cholera	diabetes	heart disease
influenza	malaria	measles	melanoma	multiple sclerosis (MS)
osteoporosis	poliomyelitis	rabies	syphilis	whooping cough

Infectious disease	Non-infectious disease

3 In Australia we are able to control many diseases that are major problems in some other countries. We have good hygiene measures, access to medicines and ways of boosting our immunity. Write the words listed in the box into the table so that they correspond to the correct health measure.

antibiotics	anti-malarial drugs	antiseptic creams	anti-viral drugs	cold-sore cream
disinfectants	exercise	hand washing	insecticides	pasteurisation
quarantine	refrigeration	using tissues	vaccination	washing food

Hygiene measures	
Medicines	
Immunity	

8.4 Immunisation

Science as a human endeavour

FOUNDATION	STANDARD	ADVANCED

Two success stories

The principles of immunisation are the same for viruses and bacteria. The two diseases below are viral diseases.

Smallpox

In the 18th century, smallpox was a disease that killed many people. In 1788 an epidemic of smallpox spread through the part of England where Edward Jenner (1749–1823) lived. He was a doctor and he noticed that people who caught cowpox, did not get smallpox. Cowpox is a similar but milder disease to smallpox.

In 1796, Jenner carried out an experiment on one of his young patients, eight-year-old James Phipps. Jenner made two small cuts on James's arm and rubbed into them a small amount of pus from a cowpox infection. James developed the slight fever that is normal for a cowpox infection. He soon recovered. A few weeks later, Jenner again made small cuts on James's arm. This time pus from a smallpox infection was rubbed into them. James remained healthy—he was immune to smallpox. Vaccination as a treatment to prevent disease had begun.

Within a few years, vaccination against smallpox was widespread. However, nearly 200 years later, about 2 million people still died from the disease each year. In the 1960s the World Health Organization began a worldwide vaccination campaign to eradicate smallpox. In December 1979 the disease was declared extinct in nature.

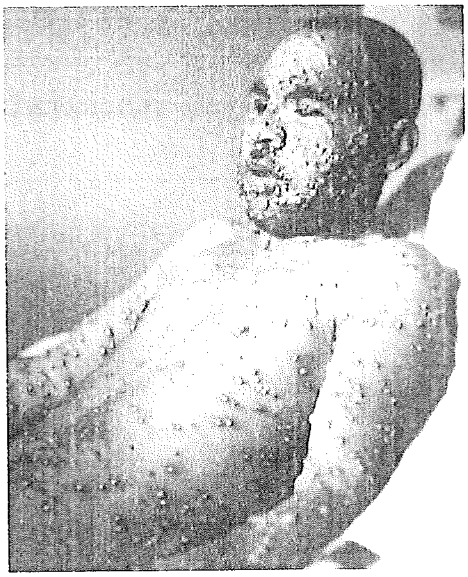

Figure 8.4.1 Smallpox was a viral disease that caused raised fluid-filled blisters. Estimates suggest that smallpox was responsible for between 300 and 500 million deaths during the 20th century.

Polio

Polio is a viral disease that causes muscle wasting, paralysis and sometimes death. In the early 1950s an American researcher, Doctor Jonas Salk (1914–1955), developed the first successful polio vaccine. To create the vaccine, Salk grew a live virus in the laboratory and then 'killed' it by adding the chemical formaldehyde. In 1954 there was nationwide testing of the vaccine on hundreds of thousands of schoolchildren. Unfortunately, one batch of the vaccine had not been made correctly and some children became ill, and a few died. Once the manufacturing process was improved, the vaccine successfully reduced the number of cases of polio.

In 1958, after 20 years of research, a Polish–American doctor, Albert Sabin (1906–1993), tested an alternative polio vaccine. Sabin's vaccine used weakened viruses and could be taken by mouth rather than by injection. It became the more popular vaccine because it was cheaper to manufacture and easier to administer (give).

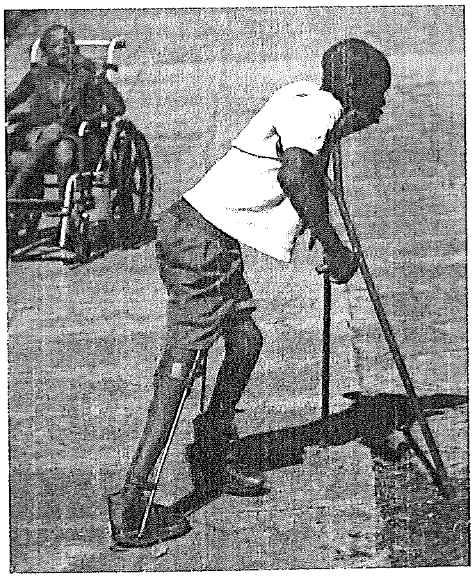

Figure 8.4.2 This boy has been crippled by polio

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8.4 Immunisation

The two vaccines between them have eradicated polio from most countries in the world.

1 Calculate the number of years between the vaccination process being tested by Jenner and the eradication of smallpox.

2 Explain the term eradication.

3 Create a flow diagram of the process Jenner used to make James Phipps immune to smallpc

4 What dangers do you think were associated with Jenner's experiment?

5 Look at Figure 8.4.2. The muscles of the boy's legs are wasted. What do you think 'wasting of muscles' means and what do you think causes it?

6 **(a)** Why did the first trials of the Salk polio vaccine cause problems?

(b) What was done to prevent these problems happening again?

7 The two vaccines for polio were made in different ways. Describe how each vaccine was made.

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RATE MY UNDERSTANDING
Shade the face that shows your rating

8.5 Viruses

Science understanding

FOUNDATION	STANDARD	ADVANCED

1 Choose one of the viruses in the box below to research. Highlight your chosen virus.

dengue	ebola	measles	mumps	rubella	zika virus

2 Research the selected virus and complete the boxes below. Information can be presented in written and/or visual form.

(a) The main countries affected:

(b) The symptoms of the virus:

(c) The effects of the virus (on people and around the world):

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(d) The life cycle of the virus. Draw it in the space below.

(e) The people most at risk of infection:

(f) Treatments for the virus:

(g) Ways to reduce the spread of the virus:

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RATE MY UNDERSTANDING
Shade the face that shows your rating

8.6 Swine flu pandemic

Science inquiry skills

FOUNDATION	STANDARD	ADVANCED

Processing & Analysing

On 24 April 2009, the United States government reported that a flu-like illness had reached the United States, with seven people confirmed as having influenza H1N1. Meanwhile, in Mexico there were 12 confirmed cases. Flu is not unusual, but this strain of flu was different. While the very young and very old are normally the people worst affected by flu, H1N1 was affecting young adults. The strain of flu was called swine flu, but it is not the same as the virus detected in pigs (swine). It was a new strain. An H1N1 virus is shown in Figure 8.6.1.

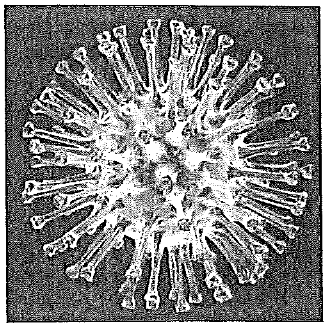

Figure 8.6.1 An H1N1 virus

confirmed (*adj*) checked as being true
strain (*n*) a variety or type

Australia recorded its first case in early May 2009. A month later the infection was declared a pandemic by the World Health Organization (WHO). A pandemic is an infectious disease that spreads rapidly over a wide area, and this infection was spreading very rapidly, as shown by the graph data in Figure 8.6.2.

The number of deaths from the disease continued to rise for the next few months (Figure 8.6.3).

On 10 August 2010, the WHO Director-General, Dr Margaret Chan, announced that the H1N1 influenza event had moved into the post-pandemic period and that although the disease would continue to be monitored there would no longer be weekly public updates.

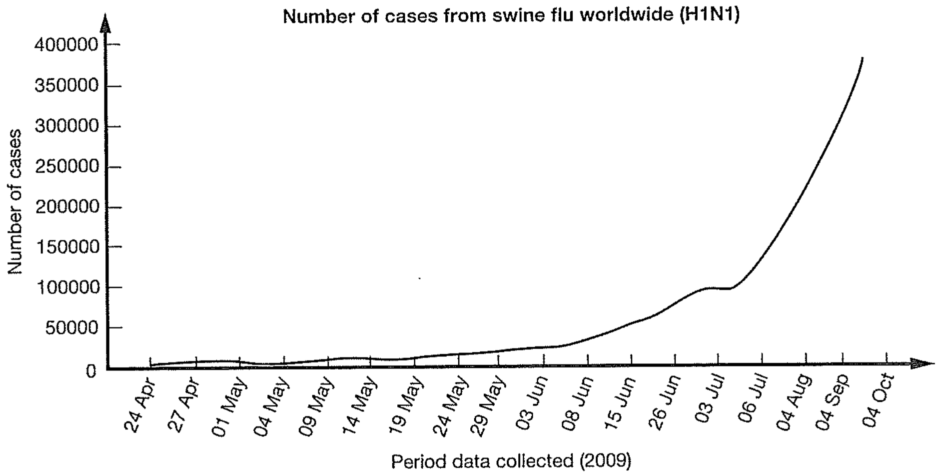

Figure 8.6.2 The number of cases of H1N1 virus increased slowly to begin with. The rate of infection then increased rapidly.

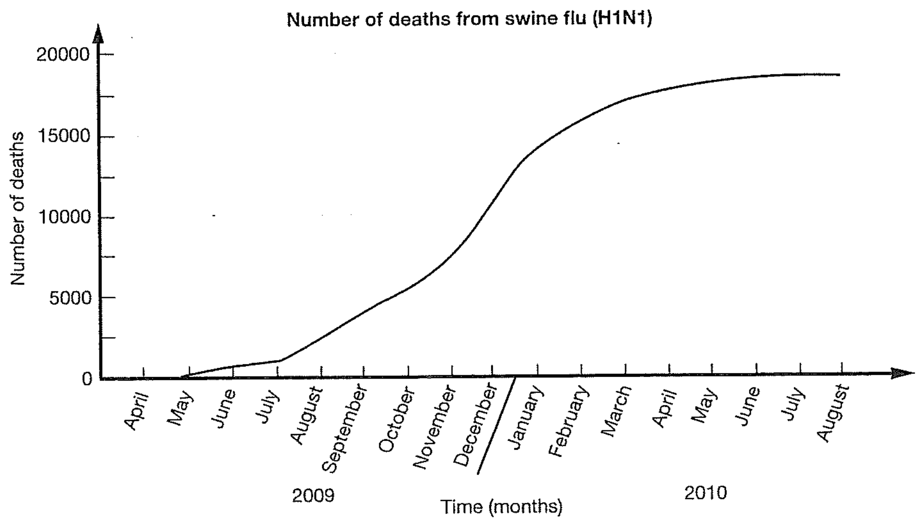

Figure 8.6.3 Number of deaths from H1N1 influenza

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8.6 Swine flu pandemic

1 What are the differences between swine flu and normal flu?

2 Provide reasons for the WHO suggesting that people avoid unnecessary travel.

3 Explain why so many people became ill even though many had been vaccinated against flu viruses.

4 Explain why the spread of swine flu was called a pandemic.

5 Refer to Figure 8.6.2 which shows the increase in the number of cases over the first few months of the H1N1 pandemic and complete the following sentences.

The number of people with swine flu, between 24 April and 15 June ______________

from 19 people to ______________, an increase of ______________ percent.

The slope of the curve between 24 April to 15 June can be described as

______________ compared with the slope between 4 August to 4 October which can be described as ______________.

The difference in the slopes of the graph at these periods means that ______________

The differences in the slope at these periods can be explained by ______________

6 Using the information in Figure 8.6.3, describe what happened to the number of deaths from H1N1 influenza from February 2010 to August 2010.

7 Deduce why the announcement was made that 'the event has moved into the post-pandemic period'.

RATE MY UNDERSTANDING
Shade the face that shows your rating

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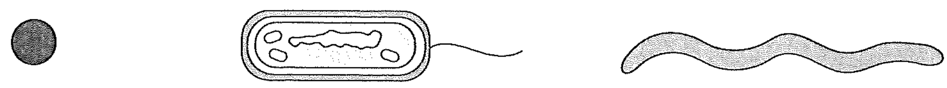
3. Name the three main shapes of bacteria shown below.



4. They can be identified by:

(a)

(b)

(c)

5. What conditions do bacteria need to grow and reproduce?

6. (a) What process do bacteria use to reproduce?

(b) Sketch it in the box provided.

(c) If one cell divided every 30 minutes, how long would it take to reach

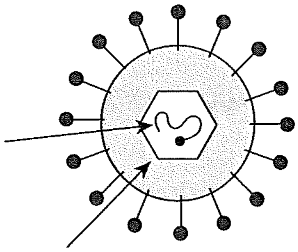
1000 cells?

100,000 cells?

1,000,000 cells?

7. Some diseases caused by bacteria include:

8. Label the diagram of a virus particle.



9. Are viruses living cells? Why or why not?

10. Viruses are described as 'obligate parasites'. Why?

11. Sketch diagrams in the boxes below to show how a virus particle can invade a cell and replicate in it. Describe what is happening in the space under each one.

12. What does 'host specific' mean?

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(b) Distinguish between what is meant by 'external' and 'internal' assisted protecti

2. Discuss three ways in which a person can assist the body in providing external protecti

(a)

(b)

(c)

3. (a) What is personal hygiene?

(b) Although avoiding all infections is not entirely possible, we can minimise the times we do become sick. What are some of the things we can do to prevent ourselves from becoming ill?

(c) What special personal hygiene is recommended for women?

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(d) What special personal hygiene is recommended for men?

4. List five ways in which transmissible diseases can be caught by people. (1 line each)

(a)

(b)

(c)

(d)

(e)

5. List some of the chemical barriers that can inhibit the growth of, or kill, pathogens.

6. Microorganisms are all around us, in the air we breathe, in the water and in the food we eat. How do the following external barriers protect us from infection?

(a) Skin

(b) Ear wax

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(e) ~~How can you avoid catching it?~~

(f) ~~Currently there is no cure, so what treatment is available?~~

7. Ebola is another viral caused disease for which there is no cure. It causes internal bleeding and patients generally die of dehydration. It is transmitted via body fluids such as blood, vomit, urine, faeces, semen and possibly sweat. In the past, it has killed 90% of those infected with it. In a recent outbreak of Ebola in West Africa, despite thousands of people catching this disease, the mortality rate was not as high as it could have been.

Suggest some reasons why the mortality rate was not as high as expected.

8. Complete the table below to show how each situation occurs.

	Natural Immunity	Artificial Immunity
Passive Immunity		
Active Immunity		

9. If you received a vaccination, what could be in it?

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10. (a) Are Australians required by law to have their children immunised against any disease?

(b) What are some diseases for which you can receive vaccinations?

(c) What are some diseases for which there are no vaccinations available?

(d) What diseases do pneumococcal, Hib and meningococcal C vaccinations prevent?

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